Benedict Leung

My research area is human-computer interaction, focusing on novel interactions with hardware like pen-based devices, eye-tracking, and brain computer interfaces. Recently, my research focuses on human-AI collaboration with Large Language Models. I have published to top-tier HCI venues, including CHI and UIST.

Education

Computer Science (MSc)
Ontario Tech University
09 / 2023 – 08 / 2025
- Led research in HCI focused on generative AI with pen-based devices for annotations.
- **Advisors**: Mariana Shimabukuro, and Christopher Collins

Computer Science (BSc Hons)
Ontario Tech University
09 / 2017 – 05 / 2022
- Minor in Mathematics
- Dean's List (Fall 2017)
- President's List (Fall 2020 - Winter 2022)
- Led research in HCI focused on interaction design for reinventing interactions for cameras
- **Advisor**: Christopher Collins

Experience

Research Associate
Ontario Tech University - Vialab
09 / 2025 – present
- Developed a language-learning application that enables users to hunt for objects to learn French words.
- Assist in publishing to ACM ETRA, research on gaze-aware interventions for distraction recovery.

Teaching Assistant
Ontario Tech University
09 / 2022 – 05 / 2025
- Programming Workshop I (Fall 2022)
- Programming Workshop II (Winter 2023)
- Software Systems Dev. & Integ. (Winter 2024, 2025)
- Human Computer Interaction (Fall 2023, 2024)

Undergraduate Student Research Fellow
Ontario Tech University - Vialab
09 / 2022 – 08 / 2023
- Engage in research activities under supervision through the Undergraduate Student Research Fellowship (USRF).
- Led research on gaze-based interfaces for question generation on videos

Undergraduate Student Researcher
Ontario Tech University - Vialab
05 / 2022 – 08 / 2022
- Work on research projects with a supervisor, funded by NSERC USRA
- Led research on crossing-based pen-selection techniques

Undergraduate Thesis
Ontario Tech University - Vialab
09 / 2021 - 04 / 2022
- Creating a touch-less camera that uses mental commands and hand gestures to interact with.
- Explores the combination of a brain-computer interface and eye-tracking glasses

Research Assistant
Ontario Tech University
07 / 2021 - 12 / 2021
- Deploy a web-based breadboard circuit-building simulator, designed for learning about digital design.

Honours and Awards

In-Course Scholarship
Ontario Tech University
09 / 2021 – 09 / 2022
- Achieved a GPA of 4.0-4.3 during the previous academic year.

NSERC USRA - $8,400
Ontario Tech University - Vialab
05 / 2022
- Engage in 16 weeks of full-time research activity under supervision.
- Awarded based on strong academic performance and relevant experience aligned with the research position.
- Funded by NSERC.

Publications

Note: *In* ***Human-Computer Interaction****, conferences are the main publication venue, offering timely dissemination and high impact. Top-tier conferences are highly selective and rely on rigorous multi-stage peer review, resulting in high-quality archival proceedings.*

Mohammed Ahmed, **Benedict Leung**, Mariana Shimabukuro, Christopher Collins, Don't Wanna Miss a Thing: Gaze-Aware Implicit Interventions for Distraction Recovery in Foreign-Language Videos. *ETRA 2026*.

Benedict Leung, Mariana Shimabukuro, Christopher Collins, AnnotateGPT: Designing Human–AI Collaboration in Pen-Based Document Annotation. *CHI 2026*.

Neel Shah, **Benedict Leung**, Mariana Shimabukuro, Ali Neshati, SwipeSense: Exploring the Feasibility of Back-of-Device Swipe Interaction Using Built-In IMU Sensors. *MobileHCI 2025*.

Benedict Leung, Mariana Shimabukuro, Matthew Chan, Christopher Collins, GazeQ-GPT: Gaze-Driven Question Generation for Personalized Learning from Short Educational Videos. *Graphics Interface 2025*.

Benedict Leung, Mariana Shimabukuro, Christopher Collins, NeuroSight: Combining Eye-Tracking and Brain-Computer Interfaces for Context-Aware Hand-Free Camera Interaction. *UIST EA 2024*.

Theses

MSc Thesis

Ontario Tech University

- Titled: Implicit Pen Annotation Assisted by Large Language Models
- Published to *CHI 2026*.
- Full thesis available here

BSc Thesis

Ontario Tech University

- Titled: Touch free camera: Mental commands and hand gestures
- Published to *UIST EA 2024*.
- Full thesis available here